

PCCSL 508 Machine Learning Lab Cycle

Experiment number	Name	Date
1	Familiarisation of Numpy and Pandas	10/7/26
2	Familiarisation of matplotlib Library	17/7/26
3	Implement simple linear Regression and Multiple Linear Regression	28/7/26
4	Implement linear regression with one variable on the California Housing dataset to predict housing prices based on a single feature (e.g., the average number of rooms per dwelling). Tasks: • Load and preprocess the dataset. • Implement linear regression using both gradient descent and the normal equation. • Evaluate the model performance using metrics such as Mean Squared Error (MSE) and R-squared. • Visualize the fitted line along with the data points.	31/7/2026 OR 5/8/26
5	Implement Ridge and Lasso regression on the Diabetes dataset. Compare the performance of these regularized models with standard linear regression. Tasks: • Load and preprocess the dataset. • Implement Ridge and Lasso regression. • Tune hyperparameters using cross-validation. • Compare performance metrics (MSE, R-squared) with standard linear regression.	7/8/26 or 12/8/26
6	Estimate the parameters of a logistic regression model using MLE and MAP on the Breast Cancer Wisconsin dataset. Compare the results and discuss the effects of regularization. Tasks: • Load and preprocess the dataset. • Implement logistic regression with MLE. • Apply MAP estimation with different regularization priors (L1 and L2 regularization). • Compare the performance and parameter estimates with MLE and MAP.	14/8/26 OR 19/8/26
7	Implement a logistic regression model to predict the likelihood of a disease using the Pima Indians Diabetes dataset. Compare the performance with and without feature scaling. Tasks: • Load and preprocess the Pima Indians Diabetes dataset. • Implement logistic regression for binary classification. • Evaluate model performance with and without feature scaling. • Analyze metrics such as accuracy, precision, recall, and F1-score.	21/8/26 OR 2/9/26

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8	Implement the K-Nearest Neighbors (KNN) algorithm for image classification using the Fashion MNIST dataset. Experiment with different values of K and analyze their impact on model performance. Tasks: • Load and preprocess the Fashion MNIST dataset. • Implement KNN for multi-class classification. • Experiment with different values of K and evaluate performance. • Discuss the impact of different K values on model accuracy and computational efficiency.	4/9/26 OR 9/9/26
9	Implement a Decision Tree classifier using the ID3 algorithm to segment customers based on their purchasing behavior using the Online Retail dataset. Analyze the tree structure and discuss the feature importance. Tasks: • Load and preprocess the Online Retail dataset. • Implement Decision Tree using the ID3 algorithm. • Visualize the decision tree and analyze feature importance. • Discuss how the tree structure helps in understanding customer behavior.	11/9/26 OR 16/9/26
10	Implement a Linear Support Vector Machine (SVM) to classify the Iris dataset. Visualize the decision boundary and discuss how the margin is determined. Tasks: • Load and preprocess the Iris dataset. • Implement a Linear SVM for binary classification (e.g., classify Setosa vs. Non-Setosa). • Visualize the decision boundary and margin. • Discuss the concept of the margin and how it influences classification.	18/9/26 OR 23/9/26
11	Implement and train a Multilayer Feed-Forward Network (MLP) on the Wine Quality dataset. Experiment with different numbers of hidden layers and neurons, and discuss how these choices affect the network's performance. Tasks: • Load and preprocess the Wine Quality dataset. • Design and implement an MLP with varying architectures (different hidden layers and neurons). • Train and evaluate the network. • Discuss the impact of architecture choices on performance.	25/9/26 OR 30/9/26

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12	Implement and compare the performance of a neural network using different activation functions (Sigmoid, ReLU, Tanh) on the MNIST dataset. Analyze how each activation function affects the training process and classification accuracy. Tasks: • Load and preprocess the MNIST dataset. • Implement neural networks using Sigmoid, ReLU, and Tanh activation functions. • Train and evaluate each network. • Compare training times, convergence, and classification accuracy.	2/10/26 OR 7/10/26
13	Implement and perform hyperparameter tuning for a neural network on the Fashion MNIST dataset. Experiment with different learning rates, batch sizes, and epochs, and discuss the impact on model performance. Tasks: • Load and preprocess the Fashion MNIST dataset. • Experiment with different hyperparameters (learning rate, batch size, epochs). • Train and evaluate the network. • Discuss how hyperparameter choices affect model performance.	9/10/26 OR 14/10/26
14	Implement and apply K-means clustering to the Digits dataset. Experiment with different numbers of clusters and evaluate the clustering results using metrics such as inertia and silhouette score. Analyze how the choice of K affects clustering performance. Tasks: • Load and preprocess the Digits dataset. • Implement K-means clustering with various numbers of clusters. • Evaluate clustering performance using inertia and silhouette score. • Analyze the impact of the number of clusters on clustering quality.	16/10/26 OR 21/10/26